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REF: VTU/BGM/BoS/ECE Syllabus/616/2024-25/ **2330**

DATE: **22 AUG 2024**

CIRCULAR

Subject: 3rd semester subjects syllabus updated for 2022 scheme ECE program regarding...

Reference:

1. The Chairpersons recommendation vide email dated 14.08.2024
2. The Hon'ble Vice-Chancellor's direction dated: 20.08.2024

This is in reference to the subject cited above, the Chairperson, Board of Studies for Electronics and Communication Engineering, VTU Belagavi has added few the syllabus of the following courses based on the feedback received.

SL NO	Suggested subject with code	In Place of	Remarks
1	Typesetting using LaTeX- BECL358E (Syllabus copy attached)	C++ Basics, BEC358C	"Students who completed Introduction to C++ Programming (BPLCK105D) in their first year are advised to enroll in either Typesetting Using LaTeX (BECL358E) or other courses from the AEC group in their third semester and should avoid taking C++ Basics (BEC358C) in the same semester."
2	Electrical Circuit Analysis -BVL306E (copy the syllabus enclosed)	Circuits and Controls - BVL306A	"Students are advised to opt for alternative courses in the 3rd semester, excluding BVL306A - Circuits and Controls, since they will be required to take BVL403 - Control System as a compulsory subject in the 4th semester."

The syllabus for these courses/subjects is enclosed with this circular for the information and reference of the stakeholders. The circular is in effect from the academic year 2024-25.

All Principals of Engineering Colleges are requested to bring the content of the circular to the attention of the concerned staff and students.

Encl: As mentioned above.

Sd/-
 REGISTRAR

To,

All the Principals of Affiliated/Constituent Engineering Colleges under the ambit of the University

Copy to

- The Hon'ble Vice-Chancellor through the secretary to VC for information
- The Registrar (Evaluation) for information and needful
- The Director, ITI,SMU,VTU Belagavi for information and needful also request to upload the circular on the University website
- The Special Officer QPDS section of VTU Belagavi for information and needful
- Office copy

20/11/2024
REGISTRAR
BE

Typesetting using LaTeX		Semester	03
Course Code	BECL358E	CIE Marks	50
Teaching Hours/Week (L: T:P: S)	0:0:2:0	SEE Marks	50
Credits	01	Exam Hours	02

Course Objectives:

- To introduce the basic syntax and semantics of the LaTeX scripting language.
- To understand the presentation of tables and figures in the document.
- To illustrate the LaTeX syntax to represent the theorems and mathematical equations.
- To make use of the libraries (Tikz, algorithm) to design the diagram and algorithms in the document.

Note: Open-source editors like TeXnikCenter, TeXstudio, Texmaker or any other equivalent editors should be used for creating documents in the Latex environment

Sl. No.	Experiments																											
01	Create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. Also include header [title of document] and footer [institute name, page number] in the document.																											
02	Create a document that displays the sample Abstract/Summary.																											
03	Create a simple title page of the VTU project Report [Use suitable Logos and text formatting].																											
04	Create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry].																											
05	Create a document that contains the following table with proper labels. <table><tr><th rowspan="2">S.No</th><th rowspan="2">USN</th><th rowspan="2">StudentName</th><th colspan="3">Marks</th></tr><tr><th>Subject1</th><th>Subject2</th><th>Subject3</th></tr><tr><td>1</td><td>4XX22XX001</td><td>Name1</td><td>89</td><td>60</td><td>90</td></tr><tr><td>2</td><td>4XX22XX002</td><td>Name2</td><td>78</td><td>45</td><td>98</td></tr><tr><td>3</td><td>4XX22XX003</td><td>Name3</td><td>67</td><td>55</td><td>59</td></tr></table>	S.No	USN	StudentName	Marks			Subject1	Subject2	Subject3	1	4XX22XX001	Name1	89	60	90	2	4XX22XX002	Name2	78	45	98	3	4XX22XX003	Name3	67	55	59
S.No	USN				StudentName	Marks																						
		Subject1	Subject2	Subject3																								
1	4XX22XX001	Name1	89	60	90																							
2	4XX22XX002	Name2	78	45	98																							
3	4XX22XX003	Name3	67	55	59																							
06	Include the side-by-side graphics/pictures/figures in the document by using the subfigure concept.																											
07	Create a document that consists of the following two mathematical equations: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$= \frac{-2 \pm \sqrt{2^2 - 4*(1)*(-8)}}{2*1}$$= \frac{-2 \pm \sqrt{4+32}}{2}$ $\varphi_{\sigma}^{\lambda} A_t = \sum_{\pi \in C_t} \text{sgn}(\pi) \varphi_{\sigma}^{\lambda} \varphi_{\pi}^{\lambda}$$= \sum_{\tau \in C_{\sigma t}} \text{sgn}(\sigma^{-1} \tau \sigma) \varphi_{\sigma}^{\lambda} \varphi_{\sigma^{-1} \tau \sigma}^{\lambda}$$= A_{\sigma t} \varphi_{\sigma}^{\lambda}$																											
08	Demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document.																											

09	Create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section.
10	Develop a simple tree diagram or hierarchical structure in the document with appropriate labels using the Tikz library.
11	Develop a document to present an algorithm using algorithm/algorithmic/algorithm2e library.
12	(A) Develop a short technical report by using suitable commands and formats of user choice. (B) Develop an article as per standard format (like IEEE Template)

Course Outcomes:

At the end of the course, the student will be able to:

- Apply basic commands to develop simple document.
- Develop script to present the tables and figures in the document.
- Illustrate script to present theorems and mathematical equations in the document.
- Develop programs to generate the complete report / article with citations and a bibliography.
- Illustrate the use of Tikz and algorithm libraries to design graphics and algorithms in the document

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each course. The student has to secure not less than 35% (18 Marks out of 50) in the semester-end examination (SEE).

Continuous Internal Evaluation (CIE):

CIE marks for the practical course are 50 Marks. The split-up of CIE marks for record/ journal and test are in the ratio 60:40.

- Each experiment is to be evaluated for conduction with an observation sheet and record write-up. Rubrics for the evaluation of the journal/write-up for hardware/software experiments are designed by the faculty who is handling the laboratory session and are made known to students at the beginning of the practical session.
- Record should contain all the specified experiments in the syllabus and each experiment write-up will be evaluated for 10 marks.
- Total marks scored by the students are scaled down to 30 marks (60% of maximum marks).
- Weightage to be given for neatness and submission of record/write-up on time.
- Department shall conduct 02 tests for 100 marks, the first test shall be conducted after the 8th week of the semester and the second test shall be conducted after the 14th week of the semester.
- In a test, test write-up, conduction of experiment, acceptable result, and procedural knowledge will carry a weightage of 60% and the rest 40% for viva-voce.
- The suitable rubrics can be designed to evaluate each student's performance and learning

ability.

- The average of 02 tests is scaled down to 20 marks (40% of the maximum marks).

The Sum of scaled-down marks scored in the report write-up/journal and marks of a test is the total CIE marks scored by the student.

SemesterEnd Evaluation (SEE):

- SEE marks for the practical course are 50 Marks.
- SEE shall be conducted jointly by the two examiners of the same institute, examiners are appointed by the University.
- All laboratory experiments are to be included for practical examination.
- Breakup of marks and the instructions printed on the cover page of the answer script to be strictly adhered to by the examiners. OR based on the course requirement evaluation rubrics shall be decided jointly by examiners.
- Students can pick one question (experiment) from the questions lot prepared by the examiners jointly.
- Evaluation of test write-up/ conduction procedure and result/viva will be conducted jointly by exam.

General rubrics suggested for SEE are mentioned here, writeup-20%, Conduction procedure and result in -60%, Viva-voce 20% of maximum marks. SEE for practical shall be evaluated for 100 marks and scored marks shall be scaled down to 50 marks (however, based on course type, rubrics shall be decided by the examiners)

Change of experiment is allowed only once and 15% of Marks allotted to the procedure part are to be made zero. The duration of SEE is 03 hours.

Rubrics suggested in Annexure-II of Regulation book

Suggested Learning Resources

- BOOK: A Short Introduction to LaTeX BY FIRUZA KARMALI (AIBARA), A book for beginners, 2019
- BOOK: Formatting Information: A Beginner's Introduction to Typesetting with LaTeX, BY PETER FLYNN, Comprehensive TeX Archive Network (2005)
- LaTeX TUTORIAL: [<https://latex-tutorial.com/tutorials/>]
- LaTeX TUTORIAL: [<https://www.javatpoint.com/latex>]

Electrical Circuit Analysis		Semester	3
Course Code	BVL306E	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	3
Examination type (SEE)	Theory		
Course objectives: 1. Apply mesh and nodal techniques to solve an electrical network. 2. Solve different problems related to Electrical circuits using Network Theorems and Two port network. 3. Familiarize with the use of Laplace transforms to solve network problems. 4. Study two port network parameters and their applications. 5. Study of RLC Series and parallel tuned circuit.			
Teaching-Learning Process (General Instructions) These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes. These are sample Strategies, which teacher can use to accelerate the attainment of the various course outcomes. • Lecture method (L) does not mean only traditional lecture method, but different type of teaching methods may be adopted to develop the outcomes. • Encourage collaborative (Group) Learning in the class. • Ask at least three HOTS(Higher order Thinking)questions in the class, which promotes critical thinking. • Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyze information rather than simply recall it. • Topics will be introduced in a multiple representation. • Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them. • Discuss how every concept can be applied to the real world-and when that's possible, it helps improve the students' understanding. • Adopt Flipped class technique by sharing the materials/Sample Videos prior to the class and have discussions on the topic in the succeeding classes.			
Module-1			
Basic Concepts: Practical sources, Source transformations, Network reduction using Star - Delta transformation, Loop and node analysis with linearly dependent and independent sources for DC and AC networks.			
Module-2			
Network Theorems: Superposition, Millman's theorems, Thevenin's and Norton's theorems, Maximum Power transfer theorem.			
Module-3			
Transient behavior and initial conditions: Behavior of circuit elements under switching condition and their Representation, evaluation of initial and final conditions in RL, RC and			

RLC circuits for AC and DC excitations.
Module-4
Laplace Transformation & Applications: Solution of networks, step, ramp and impulse responses, waveform Synthesis.
Module-5
Two port network parameters: Definition of Z, Y, h and Transmission parameters, modelling with these parameters, relationship between parameters sets. Resonance: Series Resonance: Variation of Current and Voltage with Frequency, Selectivity and Bandwidth, Q-Factor, Circuit Magnification Factor, Selectivity with Variable Capacitance, Selectivity with Variable Inductance. Parallel Resonance: Selectivity and Bandwidth, Maximum Impedance Conditions with C, L and f Variable, current in Anti-Resonant Circuit, The General Case-Resistance Present in both Branches.
Course outcome (Course Skill Set) At the end of the course, the student will be able to : 1. Determine currents and voltages using source transformation/ source shifting/ mesh/ nodal analysis and reduce given network using star delta transformation. 2. Solve problems by applying Network Theorems and electrical laws to reduce circuit complexities and to arrive at feasible solutions. 3. Analyse the circuit parameters during switching transients and apply Laplace transform to solve the given network 4. Evaluate the frequency response for resonant circuits and the network parameters for two port networks

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). The student is declared as a pass in the course if he/she secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- There are 25 marks for the CIE's Assignment component and 25 for the Internal Assessment Test component.
- Each test shall be conducted for 25 marks. The first test will be administered after 40-50% of the coverage of the syllabus, and the second test will be administered after 85-90% of the coverage of the syllabus. The average of the two tests shall be scaled down to 25 marks
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The schedule for assignments shall be planned properly by the course teacher. The teacher should not conduct two assignments at the end of the semester if two assignments are planned. Each assignment shall be conducted for 25 marks. (If two assignments are conducted then the sum of the two assignments shall be scaled down to 25 marks)
- The final CIE marks of the course out of 50 will be the sum of the scale-down marks of tests and assignment/s marks.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours**).

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.

Suggested Learning Resources:**Books**

1. M.E.Van Valkenburg (2000), Network Analysis, Prentice Hall of India, 3rd edition, 2000, ISBN:9780136110958.
2. Roy Choudhury-Networks and Systems, 2nd edition, New Age International Publications, 2006, ISBN: 9788122427677

ReferenceBooks:

3. Hayt, Kemmerly and Durbin-Engineering Circuit Analysis, **TMH**7th Edition, 2010.
4. **J.David Irwin/ R.Mark Nelms-** Basic Engineering Circuit Analysis JohnWiley,8thed,2006.
5. Charles K Alexander and Mathew NO Sadiku-Fundamentals of Electric Circuits, Tata McGraw-Hill,3rd Ed,2009.